

Status	Finished
Started	Thursday, 30 October 2025, 12:01 PM
Completed	Thursday, 30 October 2025, 12:14 PM
Duration	12 mins 33 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

The following sequence

$$p_n = -\frac{1}{2^n}$$

is a particular solution to the recurrence relation

$$3 \cdot a_n - 2 \cdot a_{n-1} = \frac{1}{2^n}.$$

a) Find another solution to the recurrence relation, not equal to p_n :

$a_n =$

Your last answer was interpreted as follows:

$$\frac{-1}{2^n} + \left(\frac{2}{3}\right)^n$$

The variables found in your answer were: $[n]$

Syntax guide: Your answer should be an expression which depends on the variable n , and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n},$$

you should type **-7*(2^n+1) / (3*n)**.

b) Given the initial condition $a_0 = -2$. Compute a_n for $n = 1$:

$a_1 =$

Your last answer was interpreted as follows:

$$-\frac{7}{6}$$

Question 2

Correct

Mark 1.00 out of 1.00

Consider $G = (P_5, \cdot_5)$, the group of integers $0 < x < 5$ that are coprime to 5, and with group operation given by multiplication modulo 5.

Recall that a subgroup $H \subseteq G$ is called *non-trivial* if $H \neq \{e\}$, and *proper* if $H \neq G$.

Select every statement below which is true, and leave the false statements unchecked.

- ☐ For each element $g \in G$, the subgroup $\langle g \rangle \subsetneq G$ is a proper subgroup of G .
- ☐ The group G is infinite.
- ☐ The group G has a non-trivial and proper subgroup.
- ☒ The order of G is 4.
- ☒ For all elements $a, b \in G$, the following is true: $ab = ba$.
- ☐ None of the above

Question **3**

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 5}$, and let $H \subset G$ be the subgroup

$$H = \{[0, 0, 0, 0, 0], [0, 0, 0, 0, 1], [1, 1, 1, 1, 0], [1, 1, 1, 1, 1]\}.$$

What is the index of $H \subset G$?

$[G : H] =$

Your last answer was interpreted as follows:

8

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that reflects around the line (x=0). Compute the index of the subgroup generate by σ :

$[D_4 : \langle \sigma \rangle] =$

Your last answer was interpreted as follows:

4

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.
Tests missing this signing code will be ignored and will not count towards the final score.

Key: 377

Signing code:

Your last answer was interpreted as follows:

23659